

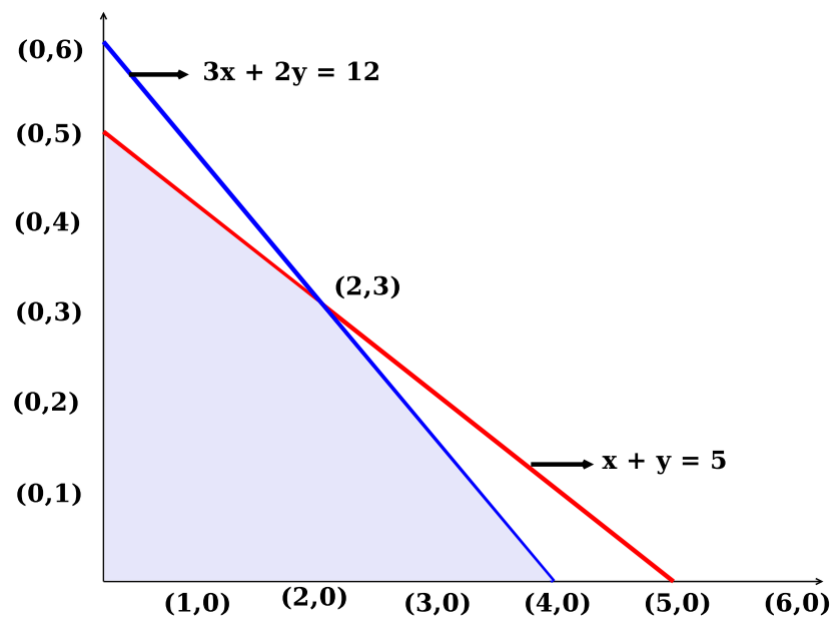
Maximise $z = 6x + 5y$
subject to

$$x + y + s = 5$$

$$3x + 2y + t = 12$$

$$x, y, s, t \geq 0$$

FIT3155: Algebraic/Tableau Simplex Side-by-side Example
Prepared by Mathew Baker based on lectures by Arun Konagurthu



Iteration 1	Non-Basic	Basic
	x, y	s, t

Write constraints in terms of non-basic variables (solve for basic variables)

$$s = 5 - x - y$$

$$t = 12 - 3x - 2y$$

Write objective function in terms of non-basic variables

$$z = 6x + 5y$$

Select variable to enter the basis.

Non-basic variable with largest positive coefficient when z is written in terms of non-basic variables $\implies x$ enters the basis.

Select variable to exit the basis.

If s exited the basis (became 0) then

$$s = 5 - x - y$$

$$0 = 5 - x - 0 \quad (s = 0, y = 0)$$

$$x = \frac{5}{1}$$

$$x = 5$$

If t exited the basis (became 0) then

$$t = 12 - 3x - 2y$$

$$0 = 12 - 3x - 2 \cdot 0 \quad (t = 0, y = 0)$$

$$x = \frac{12}{3}$$

$$x = 4$$

Selecting $x = 5$ would make $t = -3$ breaking the constraint $t \geq 0$.

So, x enters the basis ($x = 4$) and t exits the basis ($t = 0$).

Iteration 1

Write constraints in terms of non-basic variables (solve for basic variables)

	C_j	6	5	0	0		
		x	y	s	t	RHS	θ
0	s	1	1	1	0	5	
0	t	3	2	0	1	12	

Write objective function in terms of non-basic variables

	C_j	6	5	0	0		
		x	y	s	t	RHS	θ
0	s	1	1	1	0	5	
0	t	3	2	0	1	12	
		6	5	0	0	$z = 0$	

Select variable to enter the basis. The largest strictly positive (> 0) coefficient in the rewritten objective function.

	C_j	6	5	0	0		
		x	y	s	t	RHS	θ
0	s	1	1	1	0	5	
0	t	3	2	0	1	12	
		6	5	0	0	$z = 0$	

Select variable to exit the basis. The minimum strictly positive (> 0) θ value.

	C_j	6	5	0	0		
		x	y	s	t	RHS	θ
0	s	1	1	1	0	5	$\frac{5}{1} = 5$
0	t	3	2	0	1	12	$\frac{12}{3} = 4$
		6	5	0	0	$z = 0$	

Pivot occurs on red

	C_j	6	5	0	0		
		x	y	s	t	RHS	θ
0	s	1	1	1	0	5	$\frac{5}{1} = 5$
0	t	3	2	0	1	12	$\frac{12}{3} = 4$
		6	5	0	0	$z = 0$	

Iteration 2	Non-Basic	Basic
	y,t	s,x

Write constraints in terms of non-basic variables (solve for basic variables).
Solve for x :

$$t = 12 - 3x - 2y$$

$$3x = 12 - 2y - t$$

$$x = 4 - \frac{2}{3}y - \frac{1}{3}t$$

Substitute x into the equation for s :

$$s = 5 - x - y$$

$$s = 5 - (4 - \frac{2}{3}y - \frac{1}{3}t) - y$$

$$s = 1 - \frac{1}{3}y + \frac{1}{3}t$$

So

$$s = 1 - \frac{1}{3}y + \frac{1}{3}t$$

$$x = 4 - \frac{2}{3}y - \frac{1}{3}t$$

Write objective function in terms of non-basic variables

$$z = 6x + 5y$$

$$z = 6(4 - \frac{2}{3}y - \frac{1}{3}t) + 5y$$

$$z = 24 + y - 2t$$

Select variable to enter the basis.

Non-basic variable with largest positive coefficient when z is written in terms of non-basic variables $\implies y$ enters the basis.

Select variable to exit the basis.

If s exited the basis (became 0) then

$$s = 1 - \frac{1}{3}y + \frac{1}{3}t$$

$$0 = 1 - \frac{1}{3}y + \frac{1}{3} \cdot 0 \quad (s = 0, t = 0)$$

$$y = \frac{1}{\frac{1}{3}}$$

$$y = 3$$

Iteration 2

Write constraints in terms of non-basic variables (solve for basic variables).
Using row operations we make the pivot column all 0's except for the pivot row which becomes 1.

		C_j	6	5	0	0		
			x	y	s	t	RHS	θ
R_1	0	s	1	1	1	0	5	$\frac{5}{1} = 5$
R_2	0	t	3	2	0	1	12	$\frac{12}{3} = 4$
			6	5	0	0	$z = 0$	
R_3	0	s	0	$\frac{1}{3}$	1	$-\frac{1}{3}$	1	
R_4	6	x	1	$\frac{2}{3}$	0	$\frac{1}{3}$	4	

$R_4 = \frac{1}{3}R_2, R_3 = R_1 - \frac{1}{3}R_2$

Write objective function in terms of non-basic variables

		C_j	6	5	0	0		
			x	y	s	t	RHS	θ
0	s	1	1	1	0	0	5	$\frac{5}{1} = 5$
0	t	3	2	0	1	1	12	$\frac{12}{3} = 4$
		6	5	0	0	0	$z = 0$	
0	s	0	$\frac{1}{3}$	1	$-\frac{1}{3}$	1		
6	x	1	$\frac{2}{3}$	0	$\frac{1}{3}$	4		
		0	1	0	-2	$z = 24$		

$$C_1 - Z_1 = 6 - (0 \cdot 0 + 6 \cdot 1) = 0$$

$$C_2 - Z_2 = 5 - (0 \cdot \frac{1}{3} + 6 \cdot \frac{2}{3}) = 1$$

$$C_3 - Z_3 = 0 - (0 \cdot 1 + 6 \cdot 0) = 0$$

$$C_4 - Z_4 = 0 - (0 \cdot (-\frac{1}{3}) + 6 \cdot \frac{1}{3}) = -2$$

$$z = 0 \cdot 1 + 6 \cdot 4 = 24$$

If x exited the basis (became 0) then

$$x = 4 - \frac{2}{3}y - \frac{1}{3}t$$
$$0 = 4 - \frac{2}{3}y - \frac{1}{3} \cdot 0$$
$$y = \frac{4}{\frac{2}{3}}$$
$$y = 6$$

$$(x = 0, t = 0)$$

Selecting $y = 6$ would make $s = -1$ breaking the constraint $s \geq 0$.
So, y enters the basis ($x = 3$) and s exits the basis ($s = 0$).

Select variable to enter the basis.

	C_j	6	5	0	0		
		x	y	s	t	RHS	θ
0	s	1	1	1	0	5	$\frac{5}{1} = 5$
0	t	3	2	0	1	12	$\frac{12}{3} = 4$
		6	5	0	0	$z = 0$	
0	s	0	$\frac{1}{3}$	1	$-\frac{1}{3}$	1	
6	x	1	$\frac{2}{3}$	0	$\frac{1}{3}$	4	
		0	1	0	-2	$z = 24$	

Select variable to exit the basis. Pivot occurs on red

	C_j	6	5	0	0		
		x	y	s	t	RHS	θ
0	s	1	1	1	0	5	$\frac{5}{1} = 5$
0	t	3	2	0	1	12	$\frac{12}{3} = 4$
		6	5	0	0	$z = 0$	
0	s	0	$\frac{1}{3}$	1	$-\frac{1}{3}$	1	$\frac{1}{1/3} = 3$
6	x	1	$\frac{2}{3}$	0	$\frac{1}{3}$	4	$\frac{4}{2/3} = 6$
		0	1	0	-2	$z = 24$	

Iteration 3	Non-Basic	Basic
	s,t	y,x

Write constraints in terms of non-basic variables (solve for basic variables)

Solve for y

$$s = 1 - \frac{1}{3}y + \frac{1}{3}t$$

$$\frac{1}{3}y = 1 - s + \frac{1}{3}t$$

$$y = 3 - 3s + t$$

Substitute y into the equation for x :

$$x = 4 - \frac{2}{3}y - \frac{1}{3}t$$

$$x = 4 - \frac{2}{3}(3 - 3s + t) - \frac{1}{3}t$$

$$x = 2 + 2s - t$$

So

$$y = 3 - 3s + t$$

$$x = 2 + 2s - t$$

Write objective function in terms of non-basic variables

$$z = 6x + 5y$$

$$z = 6(2 + 2s - t) + 5(3 - 3s + t)$$

$$z = 27 - 3s - t$$

Select variable to enter the basis.

There are no non-basic variables with positive coefficients in the objective function.

An optimal solution has been reached.

$Z = 27$ at $x = 2, y = 3$.

Iteration 3

Write constraints in terms of non-basic variables (solve for basic variables)

		C_j	6	5	0	0		
			x	y	s	t	RHS	θ
R_1	0	s	1	1	1	0	5	$\frac{5}{1} = 5$
R_2	0	t	3	2	0	1	12	$\frac{12}{3} = 4$
			6	5	0	0	$z = 0$	
R_3	0	s	0	$\frac{1}{3}$	1	$-\frac{1}{3}$	1	$\frac{1}{1/3} = 3$
R_4	6	x	1	$\frac{2}{3}$	0	$\frac{1}{3}$	4	$\frac{4}{2/3} = 6$
			0	1	0	-2	$z = 24$	
R_5	5	y	0	1	3	-1	3	
R_6	6	x	1	0	-2	1	2	

$$R_5 = 3R_3, R_6 = R_4 - \frac{2}{3} \cdot 3R_3$$

Write objective function in terms of non-basic variables

		C_j	6	5	0	0		
			x	y	s	t	RHS	θ
0	s	1	1	1	0	0	5	$\frac{5}{1} = 5$
0	t	3	2	0	1	1	12	$\frac{12}{3} = 4$
		6	5	0	0	0	$z = 0$	
0	s	0	$\frac{1}{3}$	1	$-\frac{1}{3}$	1	1	$\frac{1}{1/3} = 3$
6	x	1	$\frac{2}{3}$	0	$\frac{1}{3}$	4	$\frac{4}{2/3} = 6$	
		0	1	0	-2	$z = 24$		
5	y	0	1	3	-1	3		
6	x	1	0	-2	1	2		
		0	0	-3	-1	$z = 27$		

$$C_1 - Z_1 = 6 - (5 \cdot 0 + 6 \cdot 1) = 0$$

$$C_2 - Z_2 = 5 - (5 \cdot 1 + 6 \cdot 0) = 0$$

$$C_3 - Z_3 = 0 - (5 \cdot 3 + 6 \cdot -2) = -3$$

$$C_4 - Z_4 = 0 - (5 \cdot -1 + 6 \cdot 1) = -1$$

$$z = 5 \cdot 3 + 6 \cdot 2 = 27$$

Select variable to enter the basis.

There are no non-basic variables with positive coefficients in the objective function.

An optimal solution has been reached.

$Z = 27$ at $x = 2, y = 3$.